

Vetting Single Sign-On SDK Implementations via Symbolic Reasoning

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Vetting Single Sign-On SDK Implementations via Symbolic Reasoning

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Abstract:

Encouraged by the rapid adoption of Single Sign-On (SSO) technology in web services, mainstream identity providers, such as Facebook and Google, have developed Software Development Kits (SDKs) to facilitate the implementation of SSO for 3rd-party application developers. These SDKs have become a critical foundation for web services. Despite its importance, little effort has been devoted to a systematic testing on the implementations of SSO SDKs, especially in the public domain. In this paper, we design and implement S3KVetter (Single-Sign-on SdK Vetter), an automated, efficient testing tool, to check the logical correctness and identify vulnerabilities of SSO SDKs. To demonstrate the efficacy of S3KVetter, we apply it to test ten popular SSO SDKs which enjoy millions of downloads by application developers. Among these carefully engineered SDKs, S3KVetter has surprisingly discovered 7 classes of logic flaws, 4 of which were previously unknown. These vulnerabilities can lead to severe consequences, ranging from the sniffing of user activities to the hijacking of user accounts.

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